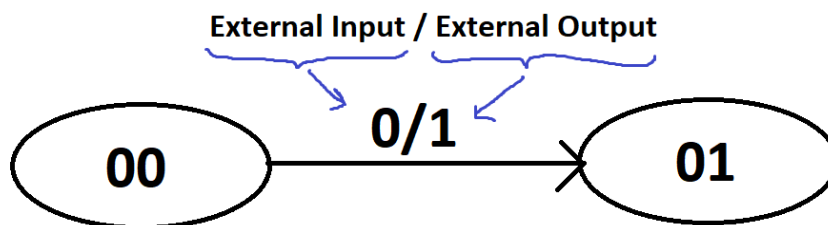


Circuit Diagram to State Diagram

Steps:

1. Find how many flip flops are there?
2. Determine:
 - a. Present States
 - b. Next States
 - c. Input (Flip flop input and external input)
 - d. Output (External Output)# a and b depends on the given flip flop.
3. Find the equation for,
 - a. All the flip flops input
 - b. External Output (If any)
4. Construct the state table according to the equation derived from step 3.
5. Extend the state table and find next state from the input values [Using Characteristics Table]
6. Draw the state Diagram



State Diagram

State Diagram to Circuit Diagram

1. Draw state table from the state diagram
2. Extend the table and figure out the inputs of the flip flop [Using Excitation Table]
3. Draw K-maps for all flip flops inputs and external outputs (if any)
4. Draw the Diagram